Module 9: Inheritance Genetics - Key Points

Learning Objectives

By the end of this module, you should be able to:

1. Distinguish gene, allele, genotype, and phenotype.
2. Use segregation to explain monohybrid inheritance.
3. Build and interpret Punnett-square probabilities.
4. Recognize inheritance patterns beyond complete dominance.
5. Use pedigree evidence to reason about traits.

Topics

- Alleles and genotype
- Mendelian patterns
- Punnett-square prediction
- Beyond simple dominance
- Pedigrees and probability

Key Terms to Know

- **Gene** - DNA sequence contributing to a trait or functional product.
- **Allele** - Variant form of a gene.
- **Genotype** - Allele combination of an organism.
- **Phenotype** - Observable trait or characteristic.
- **Dominant** - Allele effect visible with one copy in simple dominance.
- **Recessive** - Allele effect masked by dominant allele in heterozygotes.
- **Punnett square** - Tool for modeling inheritance probabilities.
- **Pedigree** - Family diagram used to infer inheritance patterns.

Core Contents

1. Alleles are gene variants that contribute to genotype and phenotype.
2. Mendelian inheritance predicts patterns when alleles segregate into gametes.
3. Punnett squares model probability, not guaranteed outcomes.
4. Incomplete dominance, codominance, polygenic traits, and environment complicate simple patterns.
5. Pedigrees use family evidence to infer inheritance patterns.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the module's companion lab (see the course schedule pairing) as the evidence check.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-10_inheritance-genetics.md`

Generated Visuals

- Module 09: Inheritance Genetics Evidence Map
- Module 09: Inheritance Genetics Reasoning Sequence
- Module 09: Inheritance Genetics Retrieval and Lab Check